# Experiment 6 Day 3: Conversion of Borneol to Camphor

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CHEM 2081

## Introduction

On Day 3 of Experiment 6, camphor was synthesized through the oxidation of borneol. In this reaction, sodium hypochloride served as the oxidizing agent while acetone and acetic acid served as the solvent. This endothermic reaction required heat through a reflux system in a heated water bath. The temperature of the system was carefully monitored to avoid production of undesirable side products and to prevent product evaporation. Thereafter, camphor was isolated by performing the washing method with methylene chloride (DCM) and water. Other reagents, such as saturated sodium bicarbonate and sodium bisulfite, were added to neutralize the organic solution. Eventually, camphor was fully isolated by removing DCM through evaporation.

## Methods and Materials

Day 3 of Experiment 6 involved converting borneol to camphor. First, 0.3 g of the borneol product from Day 2, 0.3 mL of glacial acetic acid, and 1 mL of acetone were combined in a 10 mL long-neck round-bottom flask. A reflux system with a distillation column was assembled in a water bath, which maintained a temperature between 40 and 50°C. The borneol should dissolve in the solvent at this temperature range. If not, 0.5 mL of acetone was added. Thereafter, 6 mL of aqueous sodium hypochloride solution, or bleach, was added dropwise through the top of the condenser over the duration of 35 minutes. Each dropwise addition was followed by a gentle swirl. The reflux reaction continued for 15 more minutes after the last dropwise.

Once the reflux reaction was completed, the mixture settled to cool temperature before disassembling the apparatus. The mixture was transferred to a centrifuge tube, and camphor was isolated with two portions of methylene chloride (DCM). Thereafter, about 2 mL of saturated sodium bicarbonate was added to neutralize the DCM-mixture until $CO_2$ bubbles produced. Next,

the organic, DCM layer was transferred into another centrifuge tube, where it was washed with 2 mL of 5% sodium bisulfite and 2 mL of water. To remove the residual water, anhydrous sodium sulfate was added as a drying agent. Camphor was obtained by evaporating DCM over a water bath. Temperature of the water bath was carefully monitored such that it did not exceed 50°C. Finally, the mass of the camphor product was measured.

**Results**

| | |
|---|---|
| **Borneol (g)** | 0.300 |
| **Theoretical yield of camphor (g)** | 0.296 |
| **Actual yield of camphor (g)** | 0.173 |
| **Percent yield of camphor (%)** | 58.446 |

Table 1. Starting amount of borneol; Theoretical, actual, and percent yield of camphor.

**Calculations**

**Theoretical Yield:**

$$\frac{0.300\ g\ borneol}{1}\frac{mol}{154.25g}\frac{mol\ camphor}{mol\ borneol}\frac{152.23\ g\ camphor}{mol} = 0.296\ g\ camphor$$

**Actual Yield:**

$$(watchglass + camphor) - watchglass = 39.246 - 39.073 = 0.173\ g\ camphor$$

**Percent Yield:**

$$\frac{Actual\ Yield}{Theoretical\ Yield}100\% = \frac{0.173}{0.296} * 100 = 58.446\ \%$$

**Discussion**

In this experiment, 0.173 g of camphor was obtained through the oxidation of 0.300 g of borneol. Aqueous sodium hypochloride solution was the oxidizing agent, and acetone/acetic acid were the solvents used in this reaction. The percentage yield of the product was 58.446%,

quantitatively confirming the presence of camphor. Once the endothermic reflux reaction was completed, camphor was isolated through the addition of methylene chloride (DCM). Due to its high density of 1.33 g/mL, the strongly-polar DCM captured the weakly-polar camphor into the organic layer. The resulting product solution was neutralized through the addition of saturated sodium bicarbonate and sodium bisulfite. The organic layer was further separated through the washing technique with water, which was ultimately removed with drying agent of anhydrous sodium sulfate. The organic layer, which comprised of DCM and camphor, settled to the bottom of the centrifuge tube due to its high density. Consequently, the aqueous layer with the lower density remained on top. Once the organic layer was extracted, DCM was further removed through evaporation, and camphor was completely isolated. In the separation process, there were two competing factors: solubility differences of camphor and DCM and density differences among the solvents. However, density differences proved to be more important and outweighed the significance of solubility differences. Therefore, the presence of camphor was qualitatively confirmed through density differences in the solvents during the separation process.

Nevertheless, the percentage yield of 58.446% infers that there were sources of error in the experiment that led to the loss of the camphor product. One likely source of error was the incomplete oxidation of borneol in the reflux apparatus. While the procedure instructed the reflux reaction to be completed for a total of 50 minutes, the actual process was completed in 30 minutes for time efficacy. Therefore, it is likely that the borneol was not fully oxidized by the sodium hypochloride solution, leading to a lower product yield. To avoid this mistake in the future, the experiment should be conducted as originally instructed for 50 minutes so that the maximum product yield can be obtained.

Another likely source of error may be due to the impurity of the starting material, borneol. The borneol may have contained other undesired side products that were formed from the reflux process on Day 2 of the experiment. Therefore, the lower concentration of borneol in the starting material would consequently produce lower amounts of camphor. This would directly affect the actual yield of the product and therefore lower the percentage yield. This can be avoided in future experiments by obtaining a purer form of borneol from Day 2 of the experiment.

Lastly, a source of error may be due to the incomplete separation and isolation of camphor. As mentioned, the high density of DCM allowed it to capture the camphor into the organic layer. However, the competing factor of solubility differences between highly polar DCM and weakly polar camphor may have prevented full capturing of camphor and signified that residual amounts of camphor were present in the aqueous layer. This would thereby lead to a lower percentage yield, as it was assumed that camphor settled only in the organic, DCM layer. This can be prevented in future experiments by replacing DCM with another organic solvent that is more compatible in solubility levels with camphor.

Overall, the impurity of the starting material and the incomplete separation of camphor both likely contributed to the lower percentage yield. Both errors signified a lower-than-expected camphor product, which would consequently affect the actual and percentage yield. Nevertheless, the experiment was largely successful in obtaining a reasonable amount of camphor with a percentage yield of nearly 60%.

**Reaction:**

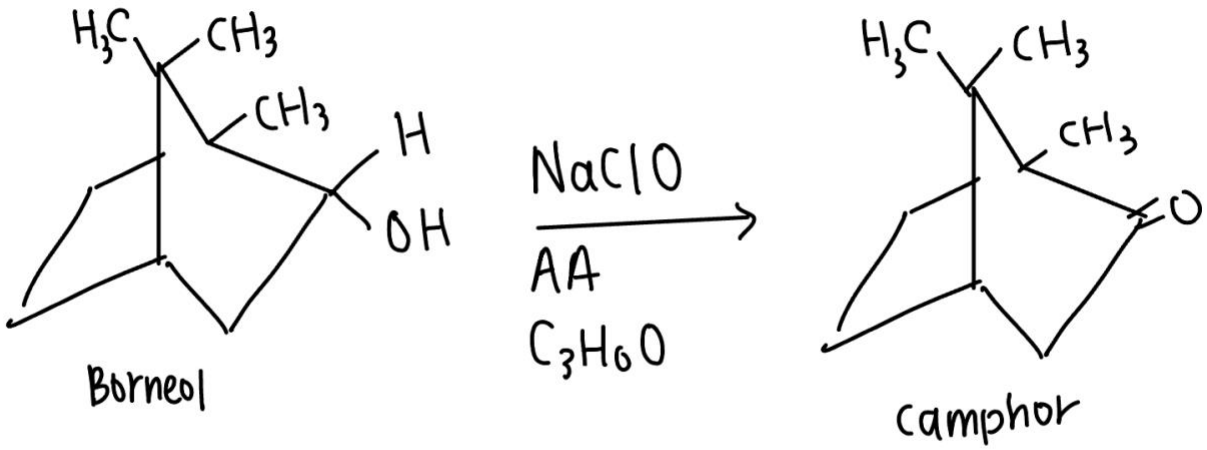



Figure 1. Oxidation of borneol to camphor.

**References**

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (AL): Auburn University.

Emma Kim

**Experiment 6 Day 3**

1.

- In long-neck 10 mL round-bottom flask, **add** racemic 0.3 g of your borneal product and 0.3 g of glacial acetic acid.
- **Dissolve** the racemic mixture in 1 mL of acetone.
- Add boiling chip.

2.

- Assemble a reflux with distillation column. Place apparatus in water bath with temperature between 40 to 50 C.
- Ensure temperature of water bath does not deviate from range.
- If borneol does not dissolve at this temp, add 0.5 mL more of acetone to mixture.

3.

- After reagents dissolve, add 6 mL of aqueous sodium hypochloride solution (bleach) dropwise for over 35 minutes through the top opening of the condenser.
- After each addition, gently swirl the mixture.
- Continue the reflux reaction for 15 more minutes after the last bleach addition.

4.

- Let mixture cool to room temp.
- Extract camphor with 2 portions (2 mL each) of methylene chloride (DCM).
  - Use centrifuge tube for this.
  - Which layer is organic? bottom

5. Neutralize combined DCM layers with ~2 mL of sat'd sodium bicarbonate. Add more bicarbonate solution if necessary until CO2 bubbles after each addition of the solution. * Also take pH
6. Wash organic solution sequentially with 2 mL of 5% sodium bisulfate and 2 mL water.

* only organic layer

1. ~~2.45+0.55~~

0.245 + 0.55 = 0.3 g

2. Borneol dissolve

↳ more acetone not needed

3.

4.

5.

6.

7. Dry organic solution with anhydrous sodium sulfate. (until freely rolling crystals of drs sodium sulfate. (until freely rolling crystals of drying agent is observed).

7.

8.

- Evaporate DCM over water bath with temp not exceeding 50C.
- Weigh collected crystals.
- Calculate percentage yield.
- Compare melting point range with that in literature.

8.

Wg: 39.073g

Wg of product: 39.246g

**Materials**

1.
   a. 0.3 g of your borneol
   b. 0.3 g of glacial acetic acid
   c. 1 mL of acetone
   d. Boiling chip
2.
   a. If necessary, 0.5 mL more of acetone
3.
   a. 6 mL of aqueous sodium hypochloride solution (bleach)
4.
   a. 2 portions (2 mL each) of methylene chloride (DCM).
5.
   a. 2 mL of sat'd sodium bicarbonate
6.
   a. 2 mL of 5% sodium bisulfate
   b. 2 mL water
7.
   a. anhydrous sodium sulfate

**Info:**

**Borneol:**
MW: 154.25
Density: 1.011 g/mL
BP: 212 C

**Acetic Acid:**
MW: 60.052
Density: 1.049 g/mL
BP: 118 C

**Acetone:**
MW: 58.08
Density: 0.784 g/mL
BP: 56 C

**Sodium hypochloride/bleach (NaClO)**
MW: 74.44
Density: 1.11 g/mL
BP: 101 C

**DCM:**
MW: 84.93
Density: 1.33 g/mL
BP: 39.6 C

**Sat'd sodium bicarbonate (NaHCO3)**
MW: 84.007
Density: 1.1 g/mL
BP: 851 C

**Water**
MW: 18.02
Density: 1 g/mL
BP: 100 C

*rmance Grade:*

If rxn is aq layer, organic should sit at bottom.

# Lab Performance

Exp 6 D3

Emma Kim

*Name:* *Section:* *Room:* *Hood:*

| | |
|---|---|
| **Wearing personal protection equipment**—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves. | ○ ○ ○ ○ ○ (5pts) |
| **Safe working habits**—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks. | ○ ○ ○ ○ ○ (5pts) |
| **Following our laboratory safety technique work procedures**—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts). | ○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts) |
| **Glassware operation**—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts). | ○(1pt) ○ (5pts) ○ (5pts) |
| **Working independently** or equal partner participation during the entire lab. | ○ ○ ○ ○ ○ (5pts) |
| **Good work behavior**—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace. | ○ ○ ○ ○ ○ (5pts) |
| **Maintain clean and organized personal space.** | ○ ○ ○ ○ ○ (5pts) |

GTA notes:

*GTA signature:* _____ *Final Performance Grade:* [ ]